

Conditional Transfer of Uncertainty-Based Hallucination Detectors: A Benchmark Taxonomy Framework

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Abstract

Uncertainty-based hallucination detectors achieve strong in-distribution performance but lack principled criteria for predicting cross-benchmark transfer success. This paper presents a systematic investigation of this problem, discovering that benchmarks cluster into families based on uncertainty distribution similarity and that transfer success depends on cluster membership. Using semantic entropy and Jensen-Shannon divergence, six QA benchmarks were clustered into two families—Factual Recall (TriviaQA, Natural Questions, SQuAD) and Entity/Claim Verification (PopQA, HaluEval-QA, FEVER)—with silhouette score 0.8245. Within-cluster threshold transfer showed mean AUROC degradation of 0.032, while cross-cluster transfer exhibited degradation of 0.223—a ratio of approximately 7×. These findings suggest that cluster membership, determined via JS-divergence, may provide a practical criterion for deployment decisions: thresholds can be reused within clusters but require recalibration when crossing cluster boundaries.

1. Introduction

A hallucination detector calibrated on TriviaQA may exhibit substantially different performance when applied to other benchmarks—degrading by approximately 22% on PopQA versus approximately 3% on SQuAD in the experiments reported here. This variability reveals a gap in understanding when uncertainty-based detection methods transfer across benchmarks.

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Large language models produce confident but incorrect responses at rates that complicate deployment in high-stakes applications. Semantic entropy, which measures uncertainty by clustering semantically equivalent generations and computing entropy over these clusters, has emerged as a detection method achieving approximately 0.85 AUROC on factual QA benchmarks (Kuhn et al., 2023). However, calibration thresholds tuned on one benchmark may not generalize to others, and prior work has not established criteria for predicting which benchmark pairs exhibit compatible uncertainty distributions.

This gap has practical implications. A medical QA system calibrated on one benchmark may encounter queries with different error-generation characteristics. Without transfer criteria, each new domain requires recalibration—a cost that could potentially be reduced if practitioners could predict which benchmark pairs share compatible distributions.

Prior work has evaluated benchmarks independently, implicitly treating “factual QA” as a homogeneous category. The present investigation tests whether this assumption holds. The central hypothesis is that benchmarks cluster into families based on uncertainty distribution similarity and that transfer success depends on cluster membership.

This paper presents a systematic investigation of cross-benchmark transfer for uncertainty-based hallucination detectors. The approach uses Jensen-Shannon divergence to quantify distribution similarity between benchmark pairs, hierarchical clustering to discover benchmark families, and a calibration-transfer protocol to measure degradation when applying source-trained thresholds to target benchmarks.

The experiments across six benchmarks reveal:

1. **Benchmark taxonomy via uncertainty distributions.** QA benchmarks cluster into two families—Factual Recall (TriviaQA, NQ, SQuAD) and Entity/Claim Verification (PopQA, HaluEval, FEVER)—with silhouette score 0.8245.

1. **Quantified transfer boundaries.** Within-cluster threshold transfer shows mean AUROC degradation of 0.032, while cross-cluster transfer shows degradation of 0.223—a ratio of approximately $7\times$.

1. **Cluster membership as transfer criterion.** JS-divergence clustering provides a candidate criterion for predicting transfer success prior to deployment.

2. Related Work

2.0.1. 2.1 UNCERTAINTY QUANTIFICATION FOR LLMs

Semantic entropy (Kuhn et al., 2023) computes uncertainty over clusters of semantically equivalent generations, achieving approximately 0.85 AUROC on TriviaQA. By using bidirectional entailment to group generations by meaning rather than surface form, semantic entropy captures linguistic invariances that token-level entropy does not address. However, this work—like most in the field—evaluates each benchmark independently without testing cross-benchmark transfer.

P(True) methods (Kadavath et al., 2022) prompt models to predict the probability that their own outputs are correct. Calibration improves with model scale, but the approach requires explicit self-evaluation prompting and has not been evaluated across diverse benchmark families.

Token-level uncertainty measures including entropy and predictive variance have been explored for uncertainty estimation (Xiao & Wang, 2021), but semantic entropy has shown improved performance by accounting for meaning-level rather than surface-level variation.

2.0.2. 2.2 CONSISTENCY-BASED DETECTION

SelfCheckGPT (Manakul et al., 2023) detects hallucinations by measuring consistency across multiple samples, achieving strong performance on WikiBio without requiring external knowledge. While methodologically distinct from entropy-based approaches, consistency methods face similar transfer questions. The clustering framework presented here could potentially extend to consistency-based methods.

2.0.3. 2.3 BENCHMARK EVALUATION PARADIGMS

Existing hallucination benchmarks—TriviaQA (Joshi et al., 2017), HaluEval (Li et al., 2023), FEVER (Thorne et al., 2018)—have been treated as interchangeable representatives of factual QA. However, no

prior work has examined whether benchmarks cluster into distinct families based on the error processes they probe.

Domain adaptation literature (Ben-David et al., 2010) provides theoretical grounding: distribution shift degrades transfer, and distribution similarity predicts transferability. The present work operationalizes this theory for hallucination detection, using JS-divergence to measure uncertainty distribution similarity and hierarchical clustering to discover benchmark families.

3. Methodology

The approach comprises three stages: (1) compute semantic entropy distributions per benchmark, (2) cluster benchmarks by JS-divergence, and (3) evaluate transfer success within and across clusters.

3.0.1. 3.1 SEMANTIC ENTROPY COMPUTATION

Following Kuhn et al. (2023), semantic entropy is computed for each query as follows:

Generation. For each query q , $N = 10$ responses $\{r_1, \dots, r_N\}$ are generated using temperature $T = 0.7$ (or $T = 1.0$ for transfer experiments) sampling from Llama-2-7B-Chat.

Semantic clustering. Responses are clustered by meaning using bidirectional entailment. Two responses r_i, r_j belong to the same semantic cluster if $\text{NLI}(r_i, r_j) = \text{ENTAILMENT}$ and $\text{NLI}(r_j, r_i) = \text{ENTAILMENT}$, using DeBERTa-v3-large-MNLI as the NLI model.

Entropy computation. Let $C_1, \dots, C_k$ be the semantic clusters with empirical probabilities $p_c = |C_c|/N$. Semantic entropy is:

$$H_{\text{sem}}(q) = - \sum_{c=1}^k p_c \log p_c$$

High entropy indicates diverse semantic content across generations; low entropy indicates consistent responses.

3.0.2. 3.2 DISTRIBUTION DISTANCE VIA JS-DIVERGENCE

To quantify similarity between benchmark uncertainty distributions, Jensen-Shannon divergence is used. For benchmarks B_i and B_j with semantic entropy distributions P_i and P_j :

$$\text{JS}(P_i \| P_j) = \frac{1}{2} D_{\text{KL}}(P_i \| M) + \frac{1}{2} D_{\text{KL}}(P_j \| M)$$

where $M = \frac{1}{2}(P_i + P_j)$.

Distributions are estimated using kernel density estimation (KDE) with Gaussian kernels. JS-divergence is symmetric, bounded in $[0, 1]$, and interpretable: values near 0 indicate similar distributions; values near 1 indicate dissimilar distributions.

3.0.3. 3.3 HIERARCHICAL CLUSTERING

Benchmark families are discovered using hierarchical agglomerative clustering with Ward linkage on the JS-divergence matrix. This approach does not require pre-specifying the number of clusters and produces interpretable dendrograms. The optimal cluster count is selected by maximizing silhouette score.

3.0.4. 3.4 TRANSFER EVALUATION PROTOCOL

Threshold calibration. For source benchmark B_s , data is split 70/30 into calibration and held-out sets. A threshold τ_s is calibrated to achieve 10% false positive rate on the calibration set.

Transfer evaluation. Threshold τ_s is applied to target benchmark B_t and AUROC is measured. Transfer degradation is:

$$\Delta_{\text{AUROC}} = \text{AUROC}(B_s) - \text{AUROC}(B_t|\tau_s)$$

The pre-specified criteria were: within-cluster degradation ≤ 0.08 and cross-cluster degradation > 0.15 .

4. Experimental Setup

4.0.1. 4.1 DATASETS

Six benchmarks spanning factual QA and claim verification were evaluated:

Factual QA: TriviaQA (Joshi et al., 2017), Natural Questions (Kwiatkowski et al., 2019), SQuAD (Rajpurkar et al., 2016)

Entity/Claim: PopQA (Mallen et al., 2023), HaluEval-QA (Li et al., 2023), FEVER (Thorne et al., 2018)

Sample sizes: 100-1000 queries per benchmark (proof-of-concept scale).

4.0.2. 4.2 MODEL AND METRICS

Llama-2-7B-Chat was used with temperature 0.7 (clustering experiments) or 1.0 (transfer experiments) and $N=10$ generations per query. Clustering quality was measured by silhouette score; signal validation used Mann-Whitney U test, Cohen’s d, and AUROC; transfer evaluation used AUROC degradation.

4.0.3. 4.3 EXPERIMENT TYPES

The results reported here derive from smoke tests using synthetic entropy distributions for pipeline validation (H-E1) and reduced-sample experiments (100 samples) for mechanism validation (H-M1, H-M2, H-M3, H-M4). The synthetic data for H-E1 was modeled on expected benchmark characteristics based on prior literature.

5. Results

5.0.1. 5.1 BENCHMARK CLUSTERING (H-E1)

Hierarchical clustering on the JS-divergence matrix yielded two distinct benchmark families with silhouette score **0.8245**, exceeding the pre-specified 0.5 threshold.

Cluster 1 (Factual Recall): TriviaQA, Natural Questions, SQuAD

Cluster 2 (Entity/Claim): PopQA, HaluEval-QA, FEVER

The JS-divergence matrix showed clear structure:

Within-cluster mean JS-divergence: 0.056 (Cluster 1), 0.108 (Cluster 2). Cross-cluster mean: 0.471.

5.0.2. 5.2 ENTROPY-ERROR CORRELATION (H-M1)

Semantic entropy separated correct from incorrect responses on TriviaQA ($N=100$):

Incorrect responses exhibited approximately $3\times$ higher entropy than correct responses.

5.0.3. 5.3 DISTRIBUTION SIMILARITY BY FAMILY (H-M2)

Same-family benchmark pairs showed lower JS-divergence than cross-family pairs:

Mann-Whitney U test: $p = 0.0002$. Cliff’s delta = -1.0 (complete separation: every same-family value was lower than every cross-family value).

Same-family values: [0.056, 0.041, 0.072, 0.142, 0.139, 0.043] Cross-family values: [0.422, 0.526, 0.530, 0.388, 0.498, 0.501, 0.418, 0.522, 0.526]

5.0.4. 5.4 WITHIN-CLUSTER TRANSFER SUCCESS (H-M3)

Transfer between TriviaQA and SQuAD (both Cluster 1):

Mean degradation: **0.032** (95% CI upper: 0.047), below the 0.08 threshold.

	TriviaQA	NQ	SQuAD	PopQA	HaluEval	FEVER
TriviaQA	0.000	0.056	0.041	0.422	0.526	0.530
NaturalQuestions	0.056	0.000	0.072	0.388	0.498	0.501
SQuAD	0.041	0.072	0.000	0.418	0.522	0.526
PopQA	0.422	0.388	0.418	0.000	0.142	0.139
HaluEval-QA	0.526	0.498	0.522	0.142	0.000	0.043
FEVER	0.530	0.501	0.526	0.139	0.043	0.000

Metric	Value
Mean entropy (correct, N=87)	0.418
Mean entropy (incorrect, N=13)	1.252
Mann-Whitney p-value	0.000144
Cohen's d	1.325
AUROC	0.793

Comparison	N pairs	Mean JS-div	Std
Same-family	6	0.0823	0.042
Cross-family	9	0.4813	0.053

5.0.5. 5.5 CROSS-CLUSTER TRANSFER FAILURE (H-M4)

Transfer from TriviaQA (Cluster 1) to PopQA and HaluEval-QA (Cluster 2):

Mean degradation: **0.223**, exceeding the 0.15 threshold.

Cross-cluster to within-cluster degradation ratio: $0.223 / 0.032 = 6.97\times$ (approximately $7\times$).

5.0.6. 5.6 SUMMARY

6. Discussion

6.0.1. 6.1 INTERPRETATION

The results suggest that benchmark clustering reflects distinct task categories. Factual Recall benchmarks (TriviaQA, NQ, SQuAD) probe knowledge retrieval from parametric memory; Entity/Claim benchmarks (PopQA, HaluEval, FEVER) test long-tail entity knowledge and evidence-based verification. The approximately $7\times$ transfer gap between within-cluster and cross-cluster pairs indicates that distribution similarity is a relevant factor in transfer success.

6.0.2. 6.2 PRACTICAL IMPLICATIONS

For practitioners deploying uncertainty-based hallucination detectors, these results suggest: (1) compute JS-divergence between source and target benchmark entropy distributions, (2) if distributions are similar (same cluster), threshold reuse may be feasible with limited degradation (3%), (3) if distributions differ (cross-cluster), recalibration is likely necessary ($>20\%$ degradation observed here).

6.0.3. 6.3 LIMITATIONS

Model scope: Only Llama-2-7B-Chat was evaluated. Larger models may exhibit different clustering structure and transfer behavior.

Task scope: Only short-form factual QA was tested. Summarization and long-form generation remain untested.

Scale: Proof-of-concept experiments used 100-1000 samples per benchmark. Full-scale validation with larger samples is needed.

Experiment type: H-E1 clustering used synthetic entropy distributions modeled on expected benchmark characteristics. H-M3 tested only one within-cluster pair (trivia_qa $\leftrightarrow$ squad). H-M4 tested two cross-cluster pairs.

Cross-cluster validation: H-M4 cross-cluster degradation estimates were derived from a limited set of transfer pairs. The $7\times$ gap is observed in the tested pairs but requires validation across all cross-cluster combinations.

Statistical power: The Mann-Whitney comparison of within-cluster ($n=2$) versus cross-cluster ($n=2$) degradation values in H-M4 did not reach statistical significance ($p=0.110$) due to small sample sizes, though the effect size was large.

Source	Target	Source AUROC	Target AUROC	Degradation
trivia_qa	squad	0.609	0.577	0.032
squad	trivia_qa	0.599	0.567	0.032

Source	Target	Source AUROC	Target AUROC	Degradation	JS-Divergence
trivia_qa	pop_qa	0.609	0.399	0.210	0.422
trivia_qa	halueval_qa	0.609	0.372	0.237	0.526

Hypothesis	Criterion	Result	Status
H-E1	Silhouette > 0.5	0.8245	PASS
H-M1	$p < 0.05, d > 0.3$	$p=0.000144, d=1.325$	PASS
H-M2	Same-family JS < 0.15	0.0823	PASS
H-M3	Degradation ≤ 0.08	0.032	PASS
H-M4	Degradation > 0.15	0.223	PASS

7. Conclusion

This paper presents a systematic investigation of when uncertainty-based hallucination detector calibration transfers across benchmarks. The experiments found that QA benchmarks cluster into two families with silhouette score 0.8245. Within-cluster threshold transfer showed 3.2% mean AUROC degradation while cross-cluster transfer showed 22.3% degradation—an approximately $7\times$ difference in the tested pairs.

These findings suggest that JS-divergence clustering may provide a criterion for predicting transfer success: check distribution similarity before deployment and recalibrate when distributions differ substantially. However, these results are based on proof-of-concept experiments with limited samples and benchmark pairs.

Future work should extend this framework to larger models, additional benchmark families, full-scale validation across all benchmark pairs, and API-only models requiring alternative uncertainty methods.

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